

Introduction to Deep Learning

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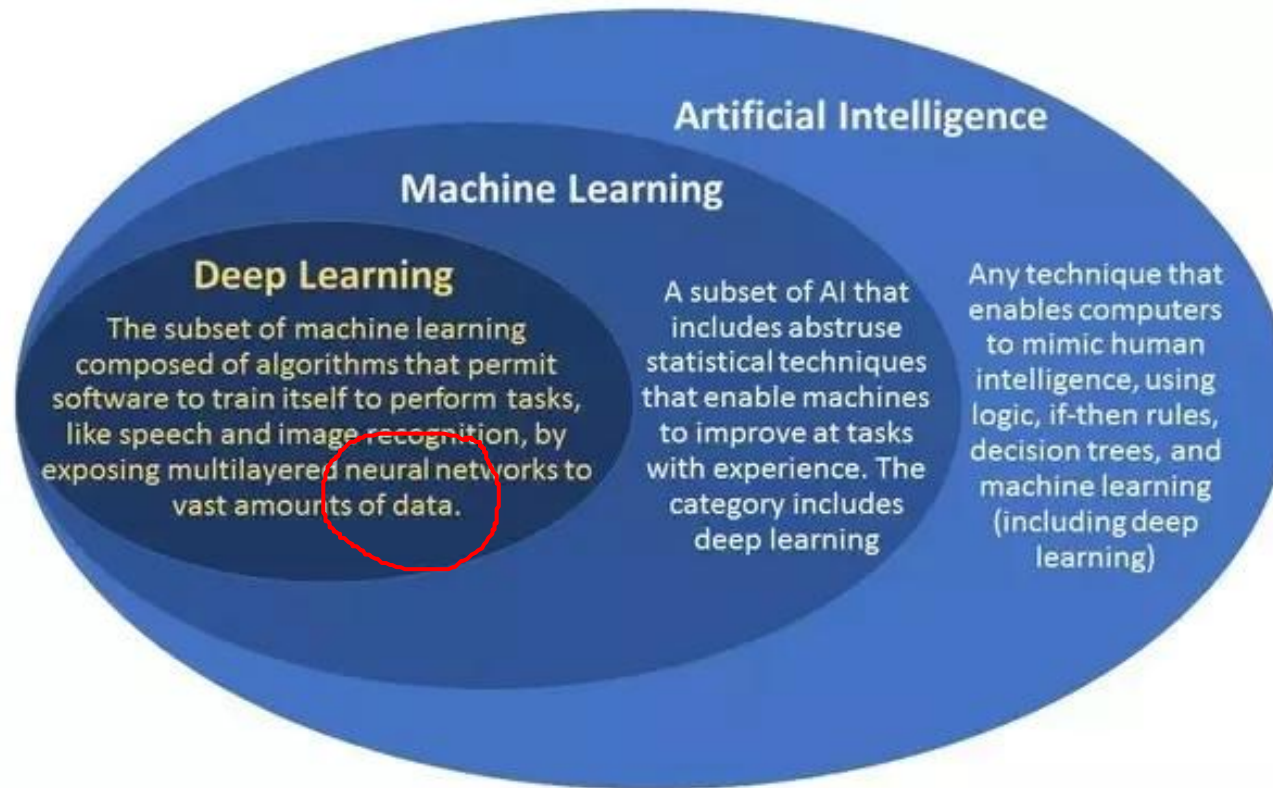
Agenda

- Part 0: Introduction
- Part 1: Perceptron
- Part 2: Structure of an ANN & Feedforward in Neural Networks
- Part 3: Tensorflow playground
- Part 4: Doubt resolution

Let us start with applications of DL

- Finance domain (**Categorical and Numerical** data) :
 - Identify the fraud detection in credit card transactions
- Healthcare domain (**Image** data):
 - Lung cancer classification of images / TB detection from chest images
- Social media(**Image** data):
 - Face recognition and tag the people
- Across the domains (**Text**):
 - Identify the potential cases of automation from historical ticket data
 - Build a chat bot

Deep learning in AI

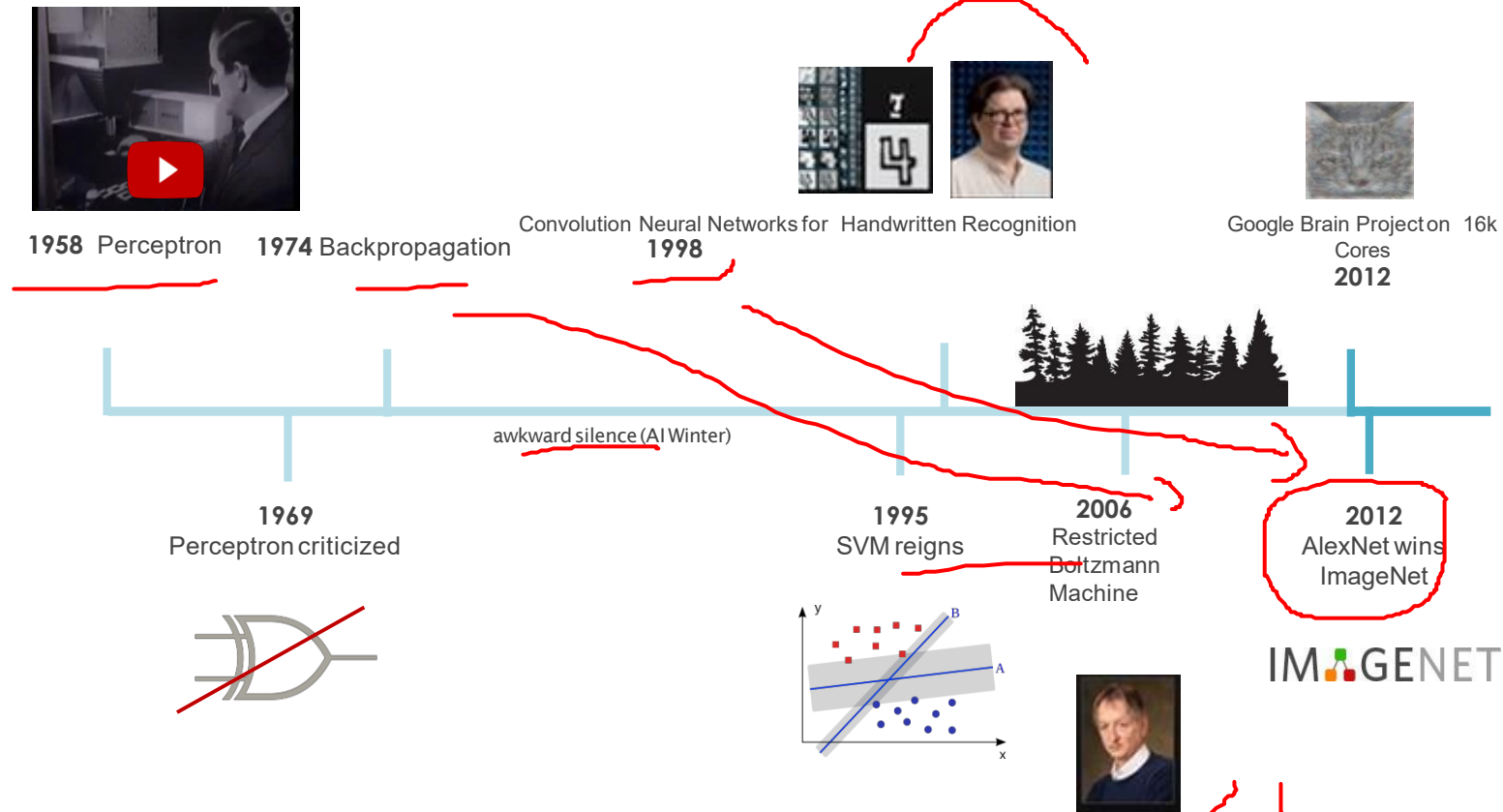


Ref:- <https://www.quora.com/What-are-the-main-differences-between-artificial-intelligence-and-machine-learning-Is-machine-learning-a-part-of-artificial-intelligence>

Brief history

NNIST 0-9

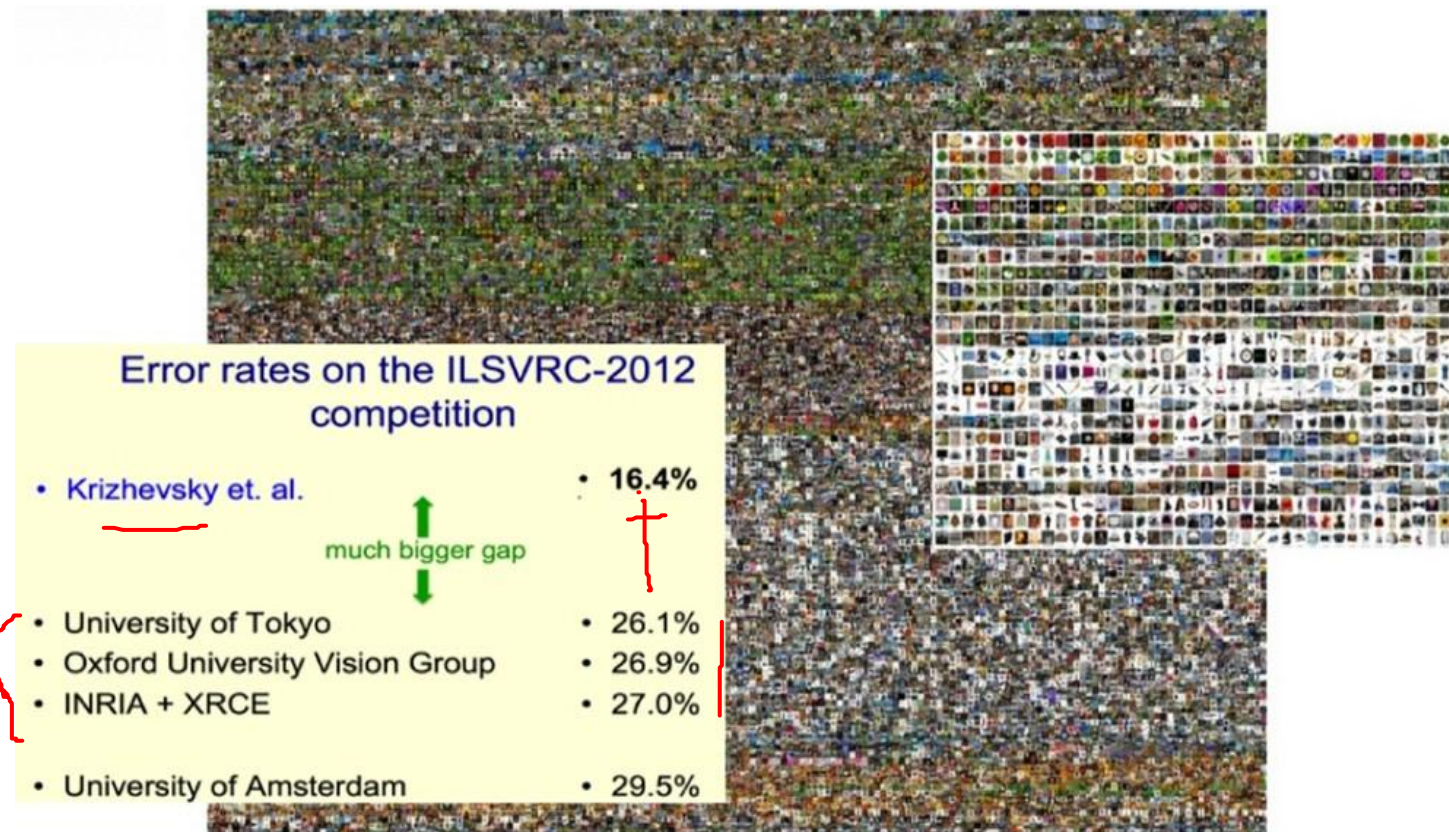
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Ref:-
<https://www.slideshare.net/mohamedloey/deep-learning-71352250>

ImageNet Challenge 2012

1.2M images with 1000 object categories



84%
374%

Why second wave?

More data from systems and sensors (IoT)

More compute power : GPU's, multi-core CPU's

Can train deep architectures

Some more applications of DL are:

Image classification, natural language processing, chat bots, personalized recommendations, prediction, anomaly detection, fraud detection, drug discovery, autonomous cars, video analytics etc...

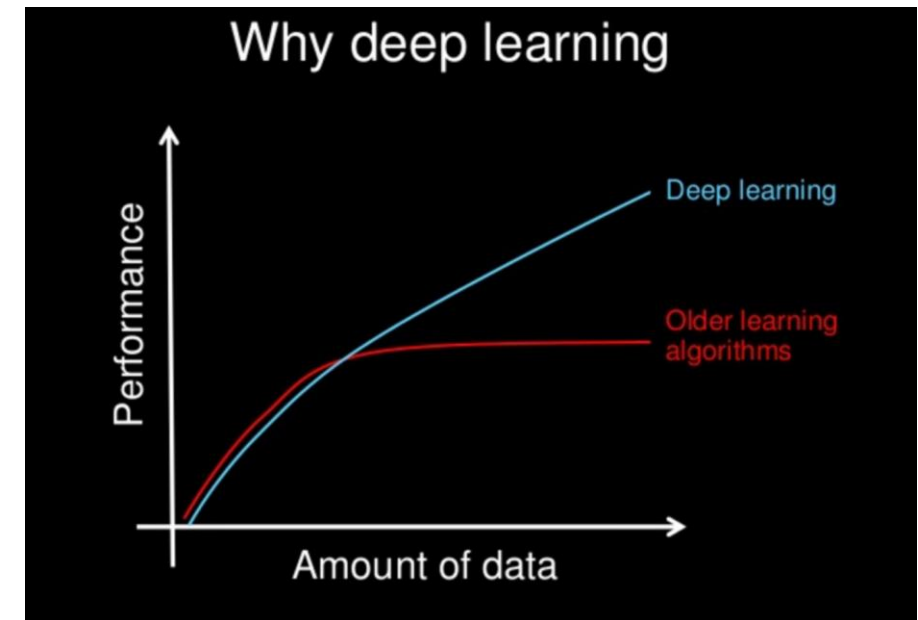
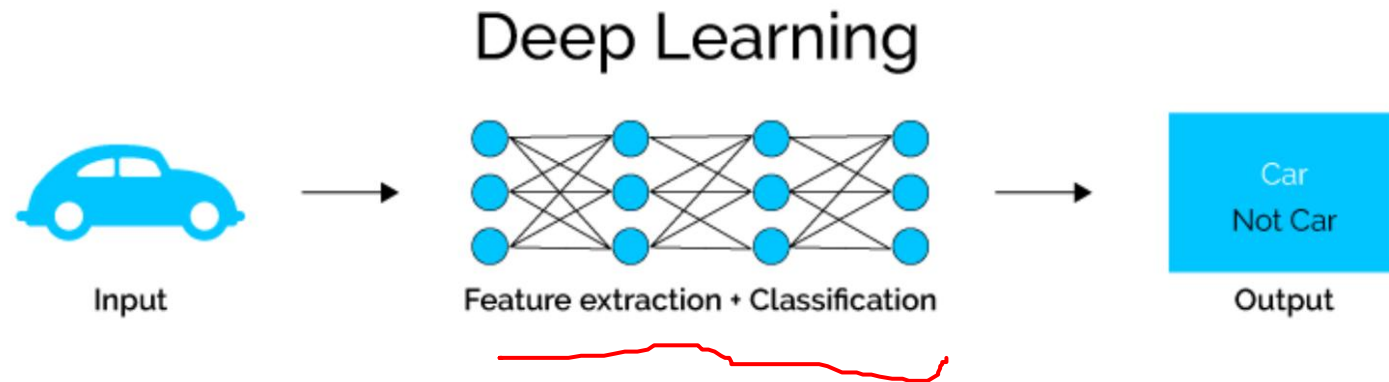
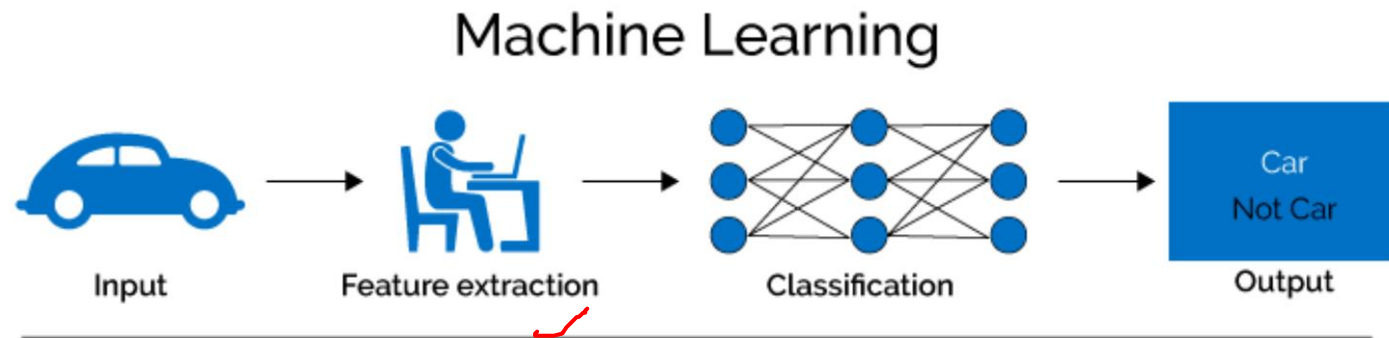


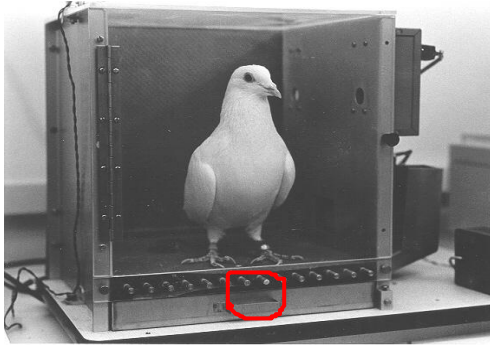
Fig Ref:- <https://www.linkedin.com/pulse/how-artificial-intelligence-revolutionizing-finance-del-toro-barba/>

Machine learning vs Deep learning



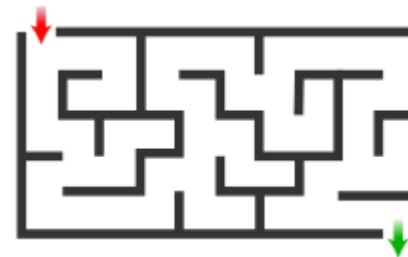
Ref:- <https://www.xenonstack.com/blog/log-analytics-with-deep-learning-and-machine-learning>

Thinking is possible even with a “small” brain



Pigeons as art experts – 85% (Watanabe *et al.* 1995)

Source: https://en.wikipedia.org/wiki/Marc_Chagall
https://en.wikipedia.org/wiki/Vincent_van_Gogh

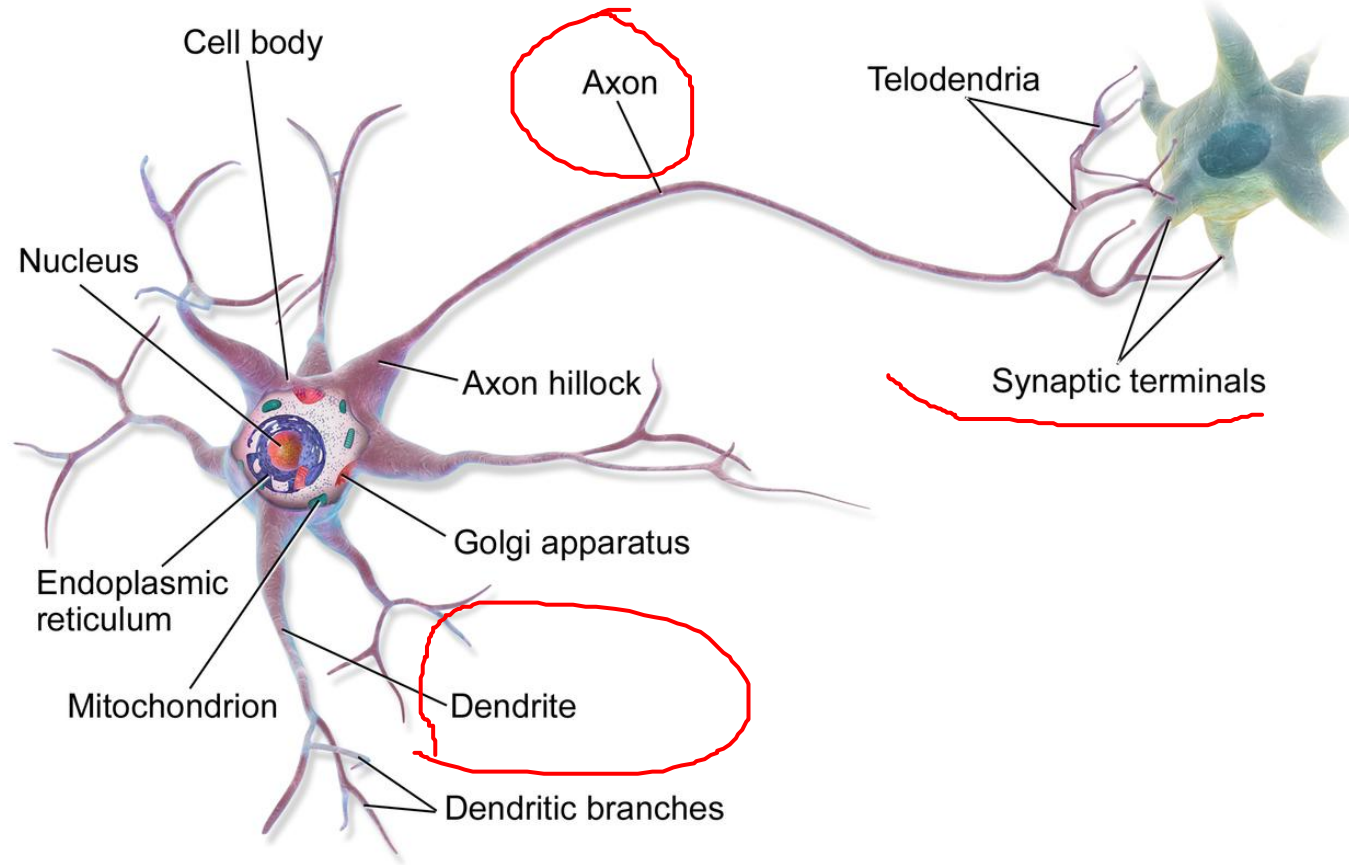


Mice trained to run mazes[1], detect drugs[1][2]

The results

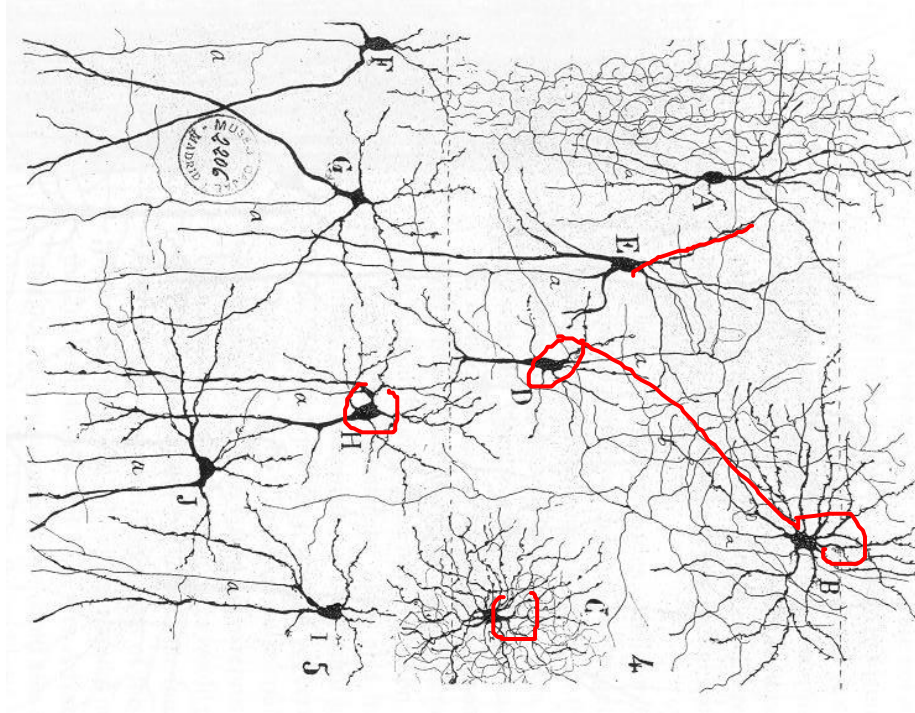
- Pigeons were able to discriminate between Van Gogh and Chagall with 95% accuracy (when presented with pictures they had been trained on)
- Discrimination still 85% successful for previously unseen paintings of the artists
- Mice can memorize mazes, odors of contraband (drugs / chemicals / explosives)

So, how does the brain work?



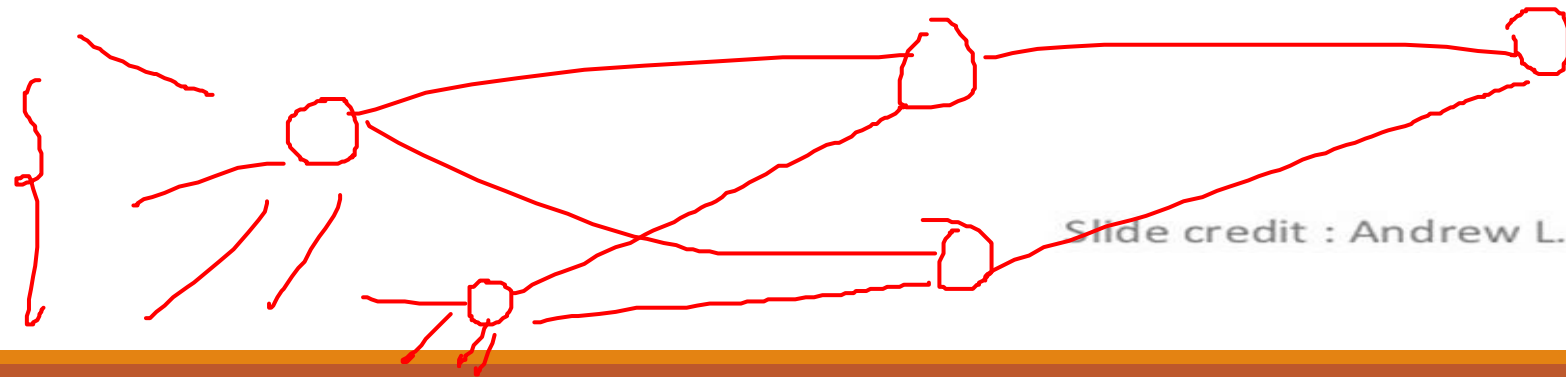
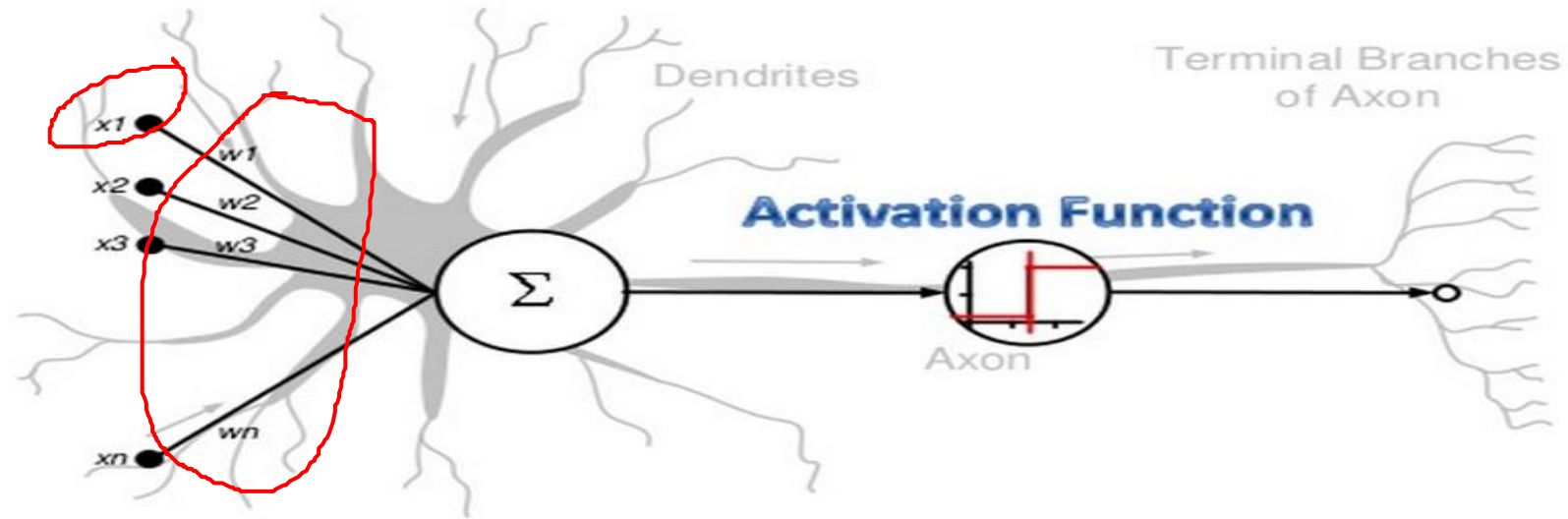
Direction of signal is along the Axon from Nucleus to synapse

Biological neural networks



- Fundamental units are termed neurons.
- Connections between neurons are synapses.
- Adult human brain consists of 100 billion neurons and 1000 trillion synaptic connections.
- Equivalent to computer with one trillion bit per second processor.

Artificial neural network



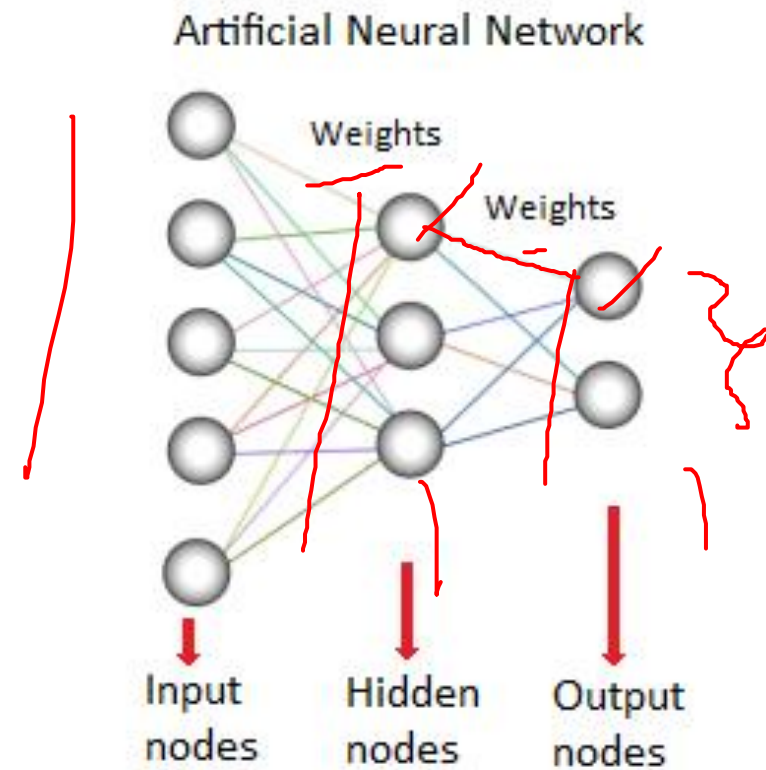
Slide credit : Andrew L. Nelson

Fundamentals

DL is a class of ML algorithms use a cascade of many layers of **nonlinear processing** units for **feature extraction** and **transformation**.

A neural net model is composed a set of **Layers**.

There are many types of layers available and each layer has many parameters. Thus we can have infinitely many different network architectures.



Artificial neural network

$$(i/p \times w + b)$$

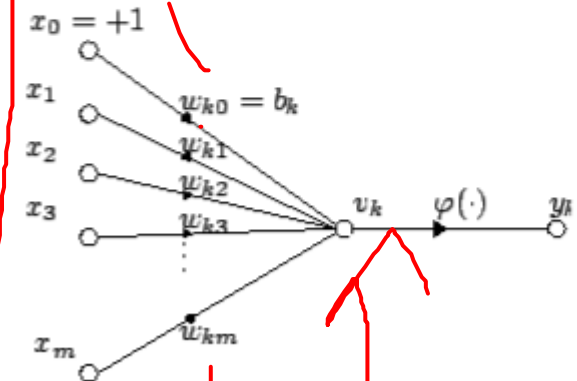
Artificial **neurons** are elementary units in an artificial neural network. The artificial neuron receives one or more **inputs** (representing dendrites) and sums them to produce an **output** (or activation) (representing a neuron's axon).

Usually the sums of each node are weighted, and the sum is passed through a non-linear function known as an **activation** function.

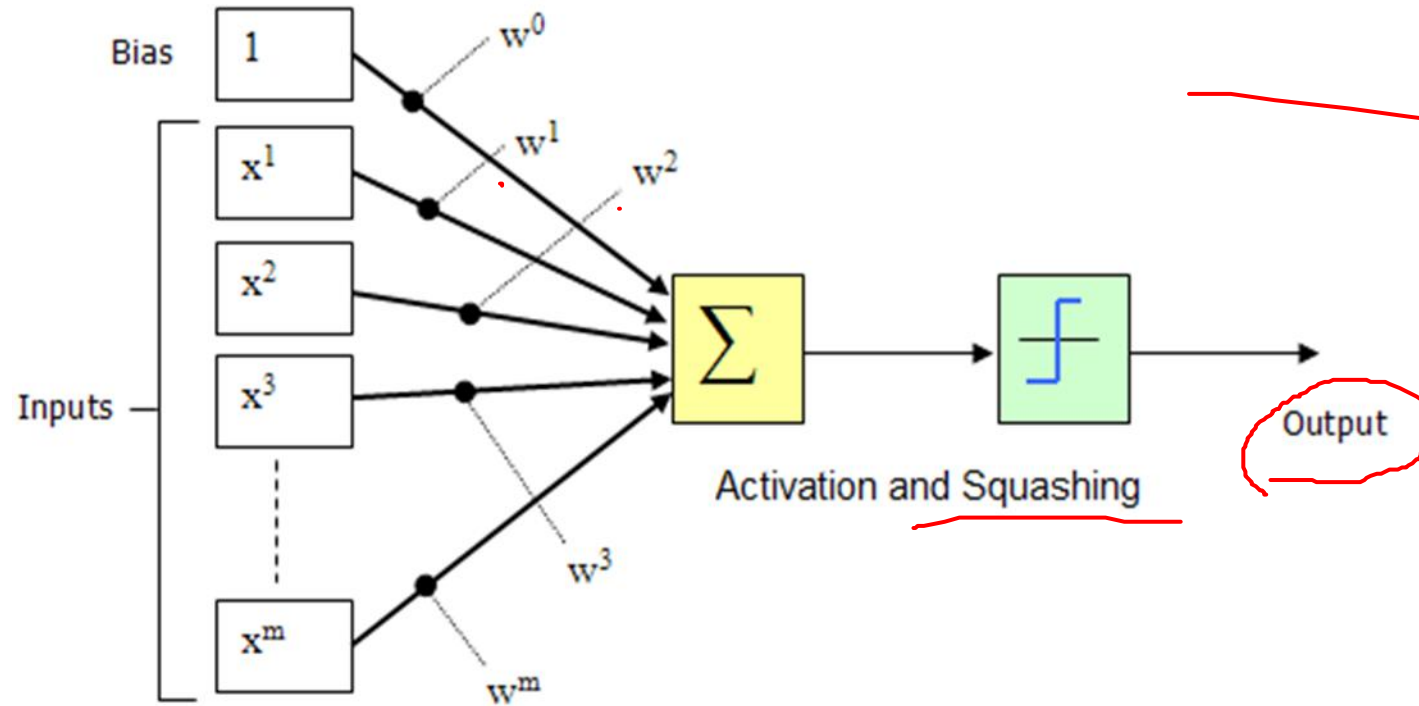
The output of the k th neuron is:

$$y_k = \varphi \left(\sum_{j=0}^m w_{kj} x_j \right)$$

Where φ (phi) is the transfer function.



Artificial neural model: Perceptron



$$\text{Output} = \{W_0 + X_1W_1 + X_2W_2 + X_3W_3 + \dots > T\}$$

$$Y = w X + b$$

$$-1 < Y < 1$$

Equation looks similar to some other model? (Linear/Logistic regression)